MIL-R-10735K
AMENDMENT-3
27 September 1979
SUPERSEDING
AMENDMENT-2
16 February 1978

MILITARY SPECIFICATION

REFRIGERATION UNITS, MECHANICAL, PANEL TYPE, GASOLINE ENGINE

AND ELECTRIC MOTOR DRIVEN, FOR SEMI-TRAILER REFRIGERATORS

This amendment forms a part of Military Specification MIL-R-10735K, dated 12 February 1975, and is approved for use by all Departments and Agenices of the Department of Defense.

PAGE 3

2.1 Add in proper sequence: "MIL-STD-1410 - Methods of Selection of Industrial Engines for End Item Application."

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3.6, line 17. After "inch" add "The plate shall be attached to the frame with vibration isolaters."

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* 3.6.4.2.3 Following the third sentence insert "The fan shall be designed for use with gasoline engines."

3.6.4.2.4, line 1. After "shaft" insert "made of stainless steel".

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* 3.6.5.5 Delete and substitute: "3.6.5.5 Liquid and liquid-moisture indicators. A liquid and liquid-moisture indicator shall be installed in the liquid refrigerant line and shall conform to MS17798-4S of MS17798. The indicator shall be installed so that it is possible to observe refrigerant condition from the front of the refrigeration unit."

3.6.5.6.1, line 12. After "engine" delete "by stalling".

FSC 4110

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3.6.5.7.1, line 8. After "minimum" delete "150" and substitute "100".

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3.6.5.16, lines 2 and 3. Delete "The condenser piping shall have flanged connections (see 3.6.5.2)."

3.6.6.1 Delete and substitute: "3.6.6.1 Gasoline engine. The gasoline engine shall be an air-cooled, 4-cycle engine capable of producing not less than 7 brake horsepower (bhp) at a speed of 1800 revolutions per minute (rpm) under continuous operation and shall be the engine manufacturer's standard commercial engine, and shall conform to MIL-STD-1410, class 1. The ignition system may be modified to meet the electromagnetic interference requirements of 3.6.10. The engine shall be opposed 2-cylinder design or a single cylinder design incorporating internal gear-driven counterweights to dynamically balance the engine. The engine shall be capable of operating on lube oil conforming to MIL-L-2104 and gasoline conforming to VV-G-1690. The engine shall be provided with a 12-volt electrical cranking system. The engine shall be furnished complete with a generator or alternator, all necessary circuit devices, controls and regulators, and mounting provisions. A fuel filter and sediment bowl shall be installed in the fuel line. All serviceable components shall be accessible and serviceable from the front of the unit without removing any other refrigeration unit components. These components shall consist of, but not be limited to, the air cleaner, spark plugs, oil dip stick, oil filler tube, oil drain, fuel filter, and carburator. The crankcase shall be provided with a permanently attached drain hose that allows the engine oil to be drained outside the unit enclosure. The gasoline engine shall be tested as specified in 4.8.17. The battery, battery cables, and battery compartment will be furnished by the semi-trailer manufacturer as components of the semi-trailer."

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* 3.6.6.1.1 Before the first sentence insert "For Type II units". In the first sentence change the capital A to small a.

Following 3.6.6.1.1 insert new paragraph as follows:

* 3.6.6.1.2 Drive. For type I and type III units a solid drive pulley with a pitch diameter of approximately 6 inches shall be furnished and mounted on the drive of the engine."

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3.6.6.3.1, line 4. Delete "commercial".

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3.6.6.3.5, last sentence. Delete and substitute "The engine hour meter shall be permanently installed in the frame of the unit."

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3.6.6.9, subparagraph (e). Add after "lever" "(when supplied)."

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3.7.1, subparagraph (c). Delete "Federal" and substitute "National".

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4.6.1.1 Delete and substitute: "4.6.1.1 Gasoline engine test. One gasoline engine out of the first ten received shall be tested as specified in 4.8.17. Thereafter, one gasoline engine selected at random out of each lot of 50 shall be tested. When less than 50 gasoline engines are submitted, one gasoline engine selected from the lot shall be tested. Failure to meet the requirements of paragraph 4.8.17 shall be cause for rejection of the lot from which the engine was selected."

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4.8.2, subparagraph (c), last sentence. Delete.

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Paragraph 4.8.14, subparagraph (2). Delete and substitute: "(2) The above mileage shall be accomplished over all courses with the trailer unloaded. The refrigeration unit shall be running, with the controls set at 0 F. for 10 hours. The unit shall be stopped and the gasoline tank filled, the oil shall be checked and oil added when necessary, the entire refrigeration unit shall be checked for loose bolts, nuts, screws, or other parts and any minor repairs shall be made. A minor repair is one which, in general, permits quick return to serviceability without extensive disassembly. At the end of 1/2 hour, the refrigeration unit shall be restarted and the road test continued. During the conduct of the road endurance test, all scheduled and unscheduled maintenance shall be performed on the refrigeration unit and the semitrailer. All operating times and the time required to perform all scheduled and unscheduled maintenance, as well as the type of maintenance, shall be recorded. Inability of the unit to operate for 3600 miles as specified shall constitute failure of this test."

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* 4.8.17 Delete and substitute: "4.8.17 Gasoline engine test. To determine compliance with 3.6.6.1 the gasoline engine shall be tested in accordance with test method 4000 of MIL-STD-1400. Failure to meet the requirements of MIL-STD-1400 shall be cause for failure of this test. Certification by the engine manufacturer that the engine meets the requirements of paragraph 3.6.6.1 may be accepted as proof of compliance."

5.1.1.1, title. Delete and substitute "Refrigeration units."

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5.2.2, lines 4 and 5. Delete "Rules".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous amendment.

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Navy - YD
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